AQ1.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Instructions:

- $(i, j)^{th}$ entry in each given matrix is a real number and scalars are also real numbers.

- If order of a matrix M is not specified, then you may assume that M is a square matrix of order 2 or 3.

4.1 Level 1:
1 point
Choose the set of correct options using Figure M2W1AQ1.

[Hint: Recall that, vector addition and scalar multiplication are done coordinatewise.]

- [x] $2A$ is the vector $(2, 4)$.
- [x] $3B$ is the vector $(6, 9)$.
- [x] $A + B$ is the vector $(3, 5)$.
- [x] $A - B$ is the vector $(-1, -1)$.

Yes, the answer is correct.
Score: 1
Accepted Answers:
$2A$ is the vector $(2, 4)$.
$3B$ is the vector $(6, 9)$.
$A + B$ is the vector $(3, 5)$.
$A - B$ is the vector $(-1, -1)$.

1 point
Let $V_1 = (1, 1)$, $V_2 = (1, 0)$, and $V_3 = (0, 1)$ be three vectors. Find out the correct set of options.

- [x] $(2, 3) = 2V_1 + 0V_2 + V_3$
- [x] $(2, 3) = 0V_1 + 2V_2 + 3V_3$
- [ ] $(2, 3) = 2V_1 + V_2 + 0V_3$
- [ ] $(2, 3) = 0V_1 + 3V_2 + 2V_3$

Yes, the answer is correct.
Score: 1
Accepted Answers:
$(2, 3) = 2V_1 + 0V_2 + V_3$
$(2, 3) = 0V_1 + 2V_2 + 3V_3$

The marks obtained by Karthika, Romy and Farzana in Quiz 1, Quiz 2 and End sem (with the maximum marks for each exam being 100) are shown in Table M2W1AQ1.

Use the above information answer questions 3, 4, and 5:

1 point
Choose the following set of correct options.

- [x] Marks obtained by Romy in Quiz 11, Quiz 22 and End sem represent a row vector.
- [x] Quiz 22 marks of Karthika, Romy and Farzana represent a column vector.
- [ ] Number of components in column vector representing Quiz 22 marks are 99.
- [x] Number of components in row vector representing Romy’s marks are 33.

Yes, the answer is correct.
Score: 1
Accepted Answers:
Marks obtained by Romy in Quiz 11, Quiz 22 and End sem represent a row vector.
Quiz 22 marks of Karthika, Romy and Farzana represent a column vector.
Number of components in row vector representing Romy’s marks are 33.

1 point
In order to improve her marks, Farzana undertook project work and succeeded in increasing her marks. Her marks became doubled for each exam. Choose the correct options.

- [x] To obtain the marks obtained by Farzana after completion of the project, scalar multiplication has to be done by 22 to the row vector representing Farzana’s marks.
- [ ] To obtain the marks obtained by Farzana after completion of the project, scalar multiplication has to be done by 11 to the row vector representing Farzana’s marks.
- [x] After completion of the project the row vector representing Farzana’s marks is $(76, 42, 7076, 42, 70)$
- [ ] After completion of the project the row vector representing Farzana’s marks is $(76, 21, 3576, 21, 35)$.
- [ ] After completion of the project the row vector representing Farzana’s marks is $(66, 82, 9066, 82, 90)$

Yes, the answer is correct.
Score: 1
Accepted Answers:
To obtain the marks obtained by Farzana after completion of the project, scalar multiplication has to be done by 22 to the row vector representing Farzana’s marks.
After completion of the project the row vector representing Farzana’s marks is $(76, 42, 7076, 42, 70)$

1 point
Following Farzana's improved marks due to her project (i.e her marks become doubled for each exam), all students were given bonus marks in Quiz 2, which is given by the column vector $(\begin{matrix}10\\12\\15\end{matrix})\left(101215\right)$.

What will be the column vector representing the final marks obtained in Quiz 2 by Karthika, Romy and Farzana?

- (•)

$\begin{pmatrix}60\\53\\57\end{pmatrix}$ 605357

○

$\begin{pmatrix}60\\53\\36\end{pmatrix}$ 605336 .

○

$\begin{pmatrix}61\\45\\53\end{pmatrix}$ 614553

○

$\begin{pmatrix}71\\57\\85\end{pmatrix}$ 715785

Yes, the answer is correct.
Score: 1
Accepted Answers:

$\begin{pmatrix}60\\53\\57\end{pmatrix}$ 605357

4.2 Level 2:

1 point

Let AA and BB be two vectors. Which of the following statements is (are) true?

- [x] $3A + 5B = 3(A + B) + [(A + B) - (A - B)]$3A + 5B = 3(A + B) + [(A + B) − (A − B)]
- [ ] $3A + 5B = 5(A + B) - [(A + B) - (A - B)]$3A + 5B = 5(A + B) − [(A + B) − (A − B)]
- [ ] $3A + 5B = 3(A + B) + [(A + B) + (A - B)]$3A + 5B = 3(A + B) + [(A + B) + (A − B)]
- [ ] $3A + 5B = 5(A + B) - [(A + B) + (A - B)]$3A + 5B = 5(A + B) − [(A + B) + (A − B)]

Partially Correct.
Score: 0.5
Accepted Answers:

$3A + 5B = 3(A + B) + [(A + B) - (A - B)]$3A + 5B = 3(A + B) + [(A + B) − (A − B)]

$3A + 5B = 5(A + B) - [(A + B) + (A - B)]$3A + 5B = 5(A + B) − [(A + B) + (A − B)]

1 point

Let $V_1 = (1, 0, 0)$V1 = (1, 0, 0), $V_2 = (0, 1, 0)$V2 = (0, 1, 0) and $V_3 = (0, 0, 1)$V3 = (0, 0, 1) be three vectors and $a, b,$ and ca, b, and c be three real numbers (scalars). Which of the following is (are) true?

- [x] $(a, b, c) = aV_1 + bV_2 + cV_3$(a, b, c) = aV1 + bV2 + cV3
- [ ]

$(a, b, c) = abV_1 + bcV_2 + caV_3$ (a, b, c) = abV1 + bcV2 + caV3

- [ ] $(a, 0, c) = aV_1 + cV_2 + 0V_3$ (a, 0, c) = aV1 + cV2 + 0V3
- [x] $(a, 0, c) = aV_1 + 0V_2 + cV_3$ (a, 0, c) = aV1 + 0V2 + cV3

Yes, the answer is correct.
Score: 1
Accepted Answers:
$(a, b, c) = aV_1 + bV_2 + cV_3$ (a, b, c) = aV1 + bV2 + cV3
$(a, 0, c) = aV_1 + 0V_2 + cV_3$ (a, 0, c) = aV1 + 0V2 + cV3

1 point
Consider vectors $A(-1, 2)$A(−1, 2) and $B(2, -2)$B(2, −2) in R^2 R2 as shown in Figure M2W1AQ2.

Choose the set of correct options.
[Hint: Recall the geometric representation of vectors, scalar multiplication and vectors addition.]

- [ ] v_1 v1 represents a scalar multiple of AA.
- [x] v_2 v2 represents a scalar multiple of AA.
- [x] v_5 v5 represents a scalar multiple of BB.
- [ ] v_1 v1 represents a scalar multiple of BB.
- [x] v_4 v4 represents a scalar multiple of $A + B$A + B.
- [ ] v_3 v3 represents a scalar multiple of $A + B$A + B.`

Yes, the answer is correct.
Score: 1
Accepted Answers:
v_2 v2 represents a scalar multiple of AA.
v_5 v5 represents a scalar multiple of BB.
v_4 v4 represents a scalar multiple of $A + B$A + B.

Let $A = (1, 1, 1)$A = (1, 1, 1) and $B = (2, -1, 4)$B = (2, −1, 4) be two vectors. Suppose $cA + 3B = (4, j, k)$cA + 3B = (4, j, k), where c, j, kc, j, k are real numbers (scalars). Find the value of cc.

-2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) -2

1 point
Let $A = (1, 1, 1)$A = (1, 1, 1) and $B = (2, -1, 4)$B = (2, −1, 4) be two vectors. Suppose $cA + 3B = (4, j, k)$cA + 3B = (4, j, k), where c, j, kc, j, k are real numbers (scalars). Find the value of $j + k$j + k.

5

Yes, the answer is correct.
Score: 1
Accepted Answers:

(Type: Numeric) 5
1 point

Check Answers
Your score is: 9.5/10